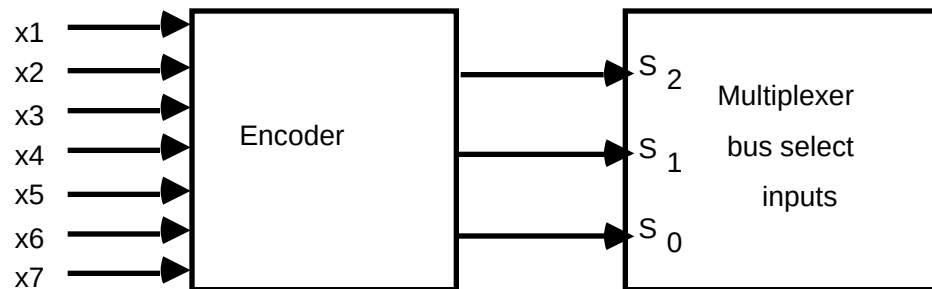


CONTROL OF COMMON BUS



x1	x2	x3	x4	x5	x6	x7	S2	S1	S0	selected register
0	0	0	0	0	0	0	0	0	0	none
1	0	0	0	0	0	0	0	0	1	AR
0	1	0	0	0	0	0	0	1	0	PC
0	0	1	0	0	0	0	0	1	1	DR
0	0	0	1	0	0	0	1	0	0	AC
0	0	0	0	1	0	0	1	0	1	IR
0	0	0	0	0	1	0	1	1	0	TR
0	0	0	0	0	0	1	1	1	1	Memory

For AR

$$\begin{array}{l} D_4T_4: PC \leftarrow AR \\ D_5T_5: PC \leftarrow AR \end{array}$$

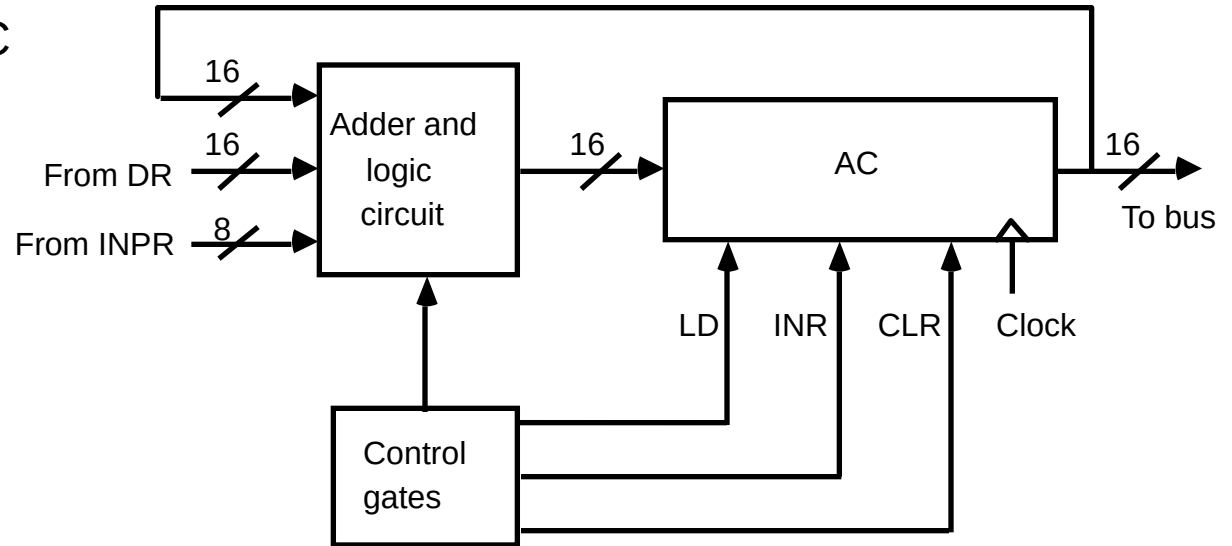


$$x1 = D_4T_4 + D_5T_5$$



DESIGN OF ACCUMULATOR LOGIC

Circuits associated with AC



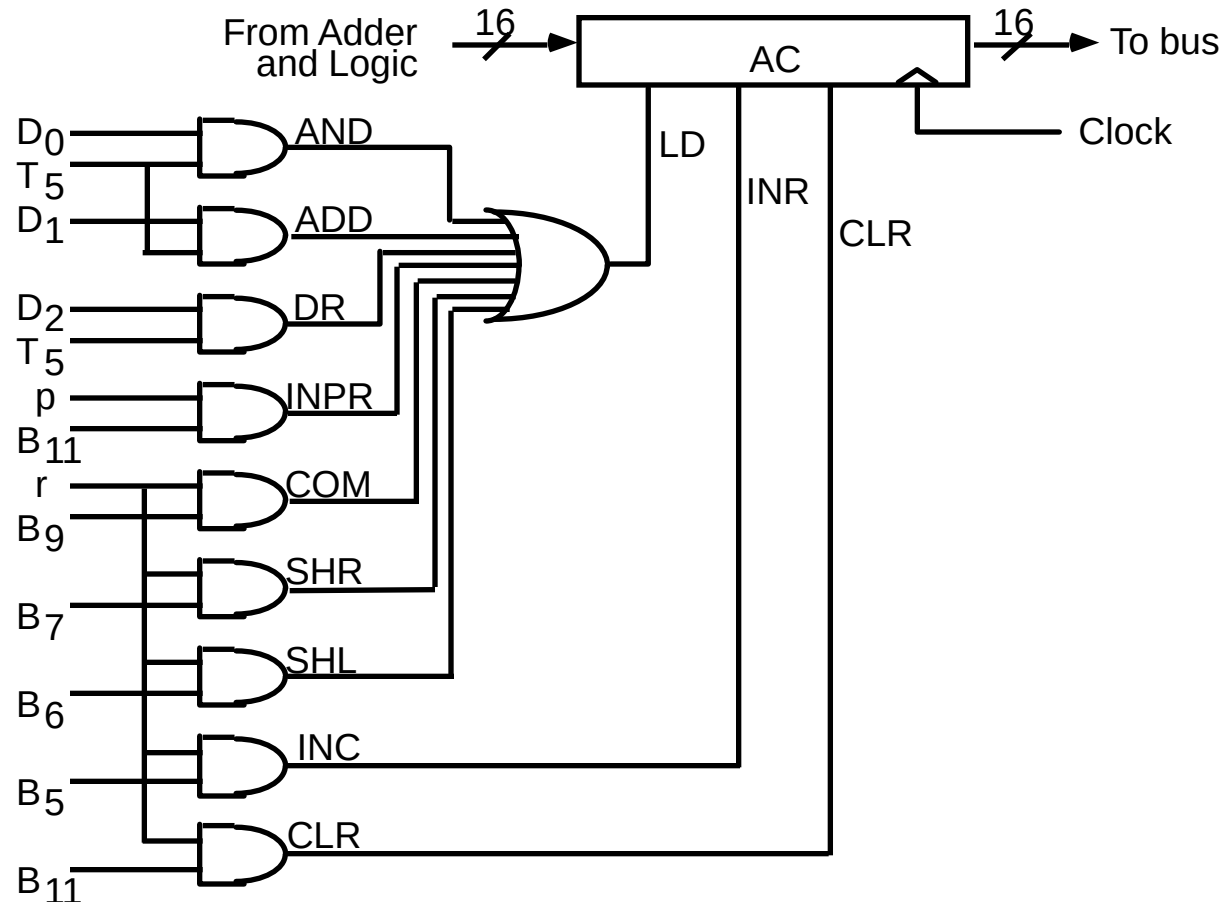
All the statements that change the content of AC

$D_0T_5:$	$AC \leftarrow AC \wedge DR$	AND with DR
$D_1T_5:$	$AC \leftarrow AC + DR$	Add with DR
$D_2T_5:$	$AC \leftarrow DR$	Transfer from DR
$pB_{11}:$	$AC(0-7) \leftarrow INPR$	Transfer from INPR
$rB_9:$	$AC \leftarrow AC'$	Complement
$rB_7:$	$AC \leftarrow shr\ AC, AC(15) \leftarrow E$	Shift right
$rB_6:$	$AC \leftarrow shl\ AC, AC(0) \leftarrow E$	Shift left
$rB_{11}:$	$AC \leftarrow 0$	Clear
$rB_5:$	$AC \leftarrow AC + 1$	Increment



CONTROL OF AC REGISTER

Gate structures for controlling the LD, INR, and CLR of AC



ALU (ADDER AND LOGIC CIRCUIT)

One stage of Adder and Logic circuit

